



Oracle Database



Performance Analysis with Panorama

Quick overview

- Peter Ramm

August 2025



About Otto Group one.O

Fusion to one.O
2025

Parent company
Otto Group

Locations
Dresden, Hamburg, Altenkunstadt, Madrid,
Málaga, Valencia, Taipei, Hyderabad

Number of employees
around 1,000

Management Board
Katrin Behrens, Dr. Stefan Borsutzky

About me

Peter Ramm



Software architekt / Team lead at One.O in Dresden

More than 30 years of history in IT-projects

Main focus:

- Development of OLTP systems based on Oracle databases
- From architecture consulting up to trouble shooting
- Performance analysis and optimization of working systems

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Agenda

- Details about the used tool “Panorama”
- Executed SQL statements with detail analysis
- Top/down analysis of session activities
- Segment statistics: Recording of characteristic values for objects (tables, indexes)

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Panorama for Oracle Databases

Tool for performance analysis of Oracle DB Free of charge (GPL3)

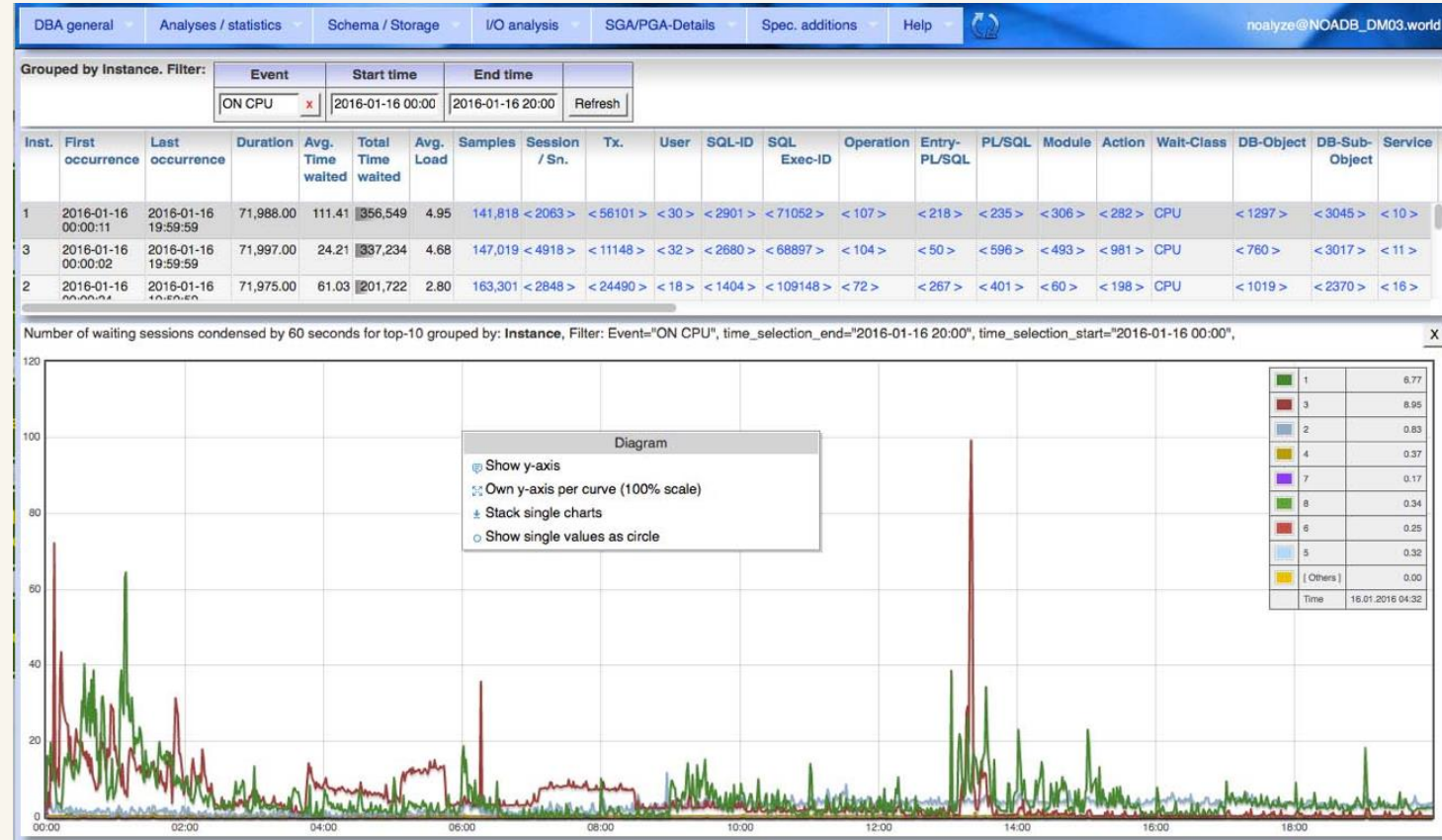
Based on dynamic performance views and
- AWR recordings
- or Panorama's own sampling
(for use with SE or without Diagnostics Pack)

Docker image or self-starting jar file

Panorama accesses your DB on a read-only basis and does not install any DB objects of its own. (except using Panorama-Sampler)

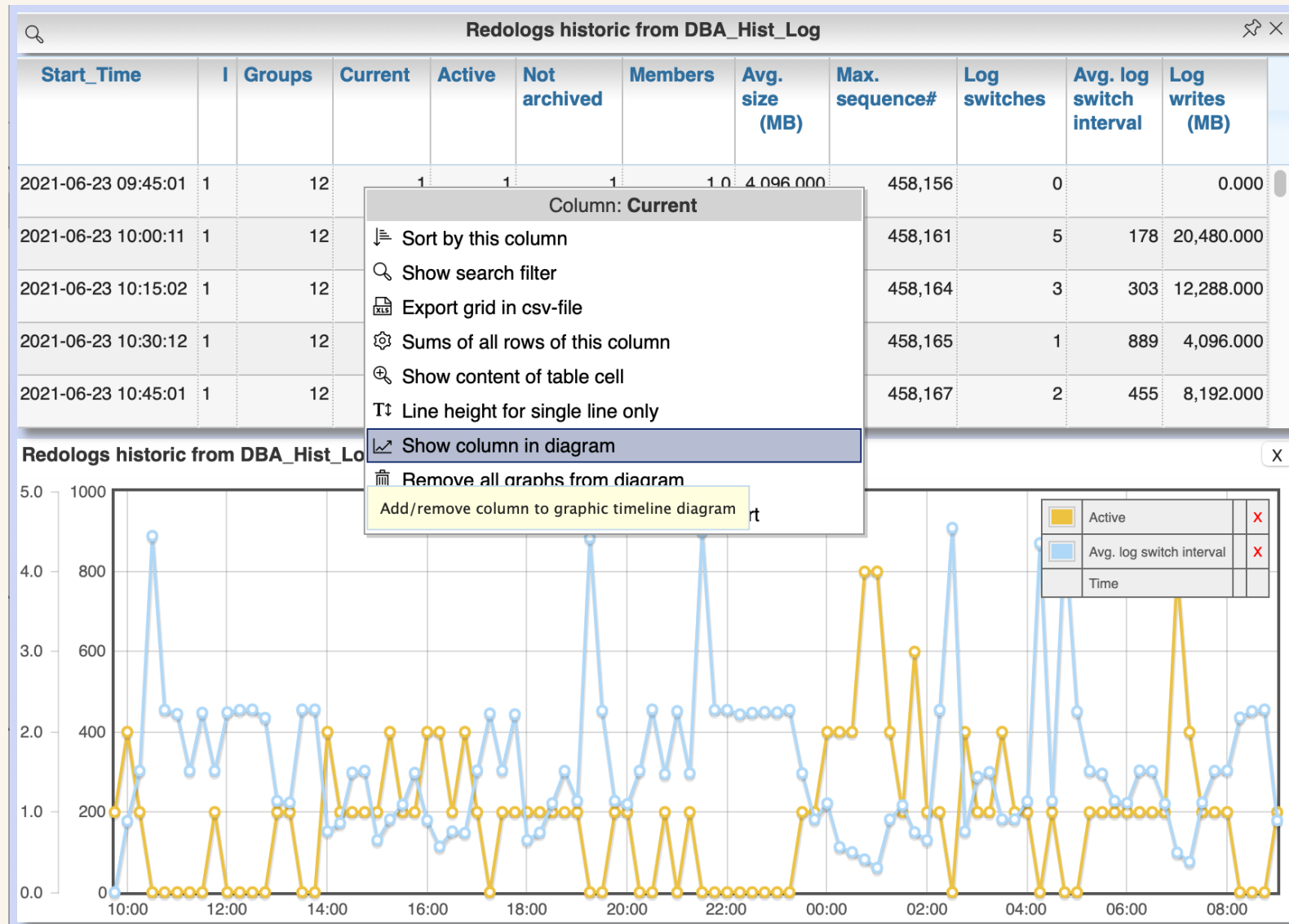
So you can test the functions without any risk.

Description of Panorama incl. download link :
Oracle performance analysis blog :
Various slides on the use of Panorama :



<https://rammpeter.github.io/panorama.html>
<https://rammpeter.blogspot.com>
<https://www.slideshare.net/PeterRamm1>

Panorama for Oracle: presentation formats



Workflow in browser page continuously downwards

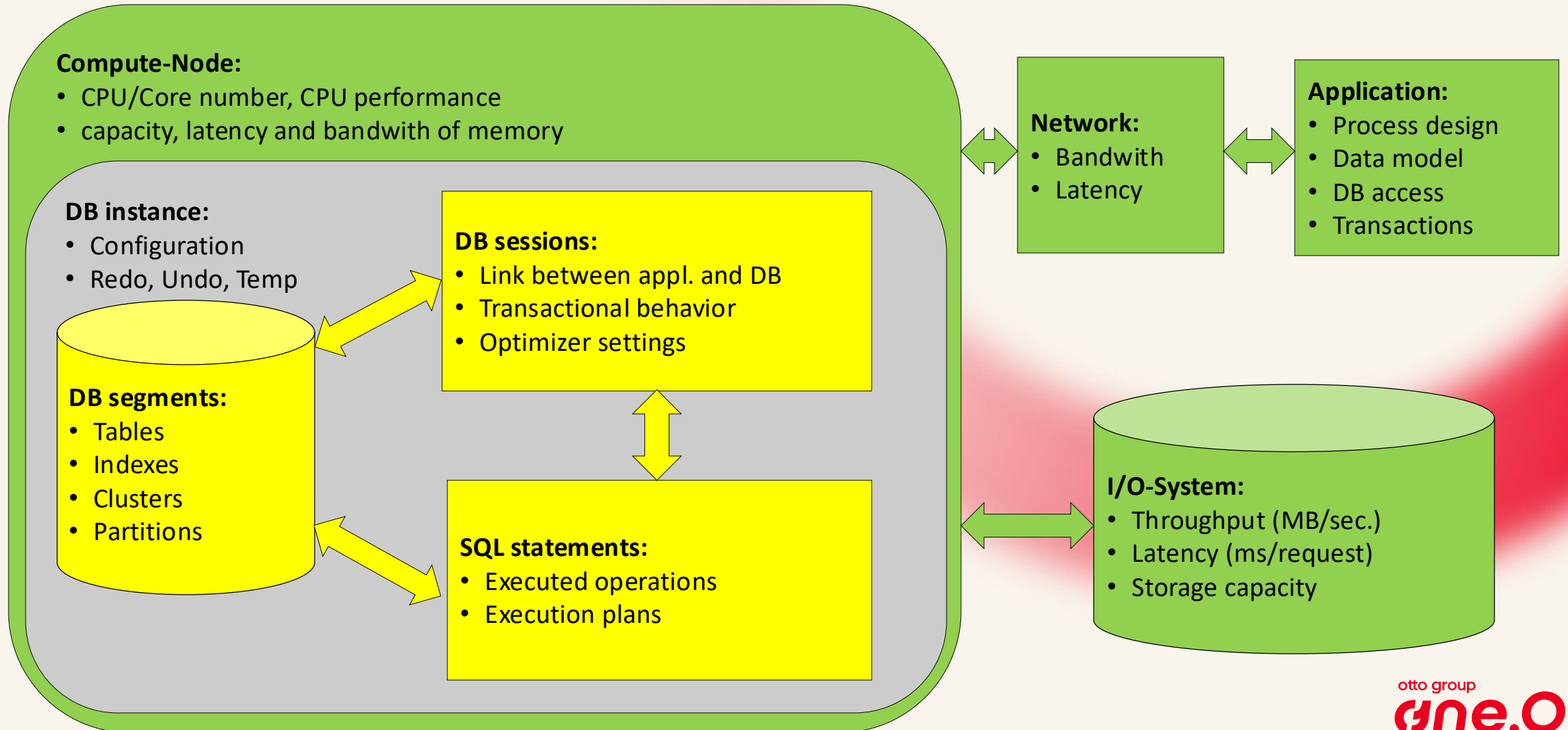
Results usually as tables

Time-related values can generally also be displayed as graphs by showing/hiding columns of the tables

Visualization in graphs via context menu (right mouse button)

Environment variable
PANORAMA_LOG_LEVEL=debug shows
executed SQLs on console

Factors influencing performance in DB use



Agenda

- Details about the used tool “Panorama”
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Executed SQLs

Menu „SGA/PGA details“ / „SQL-Area“ / ..

SQL of current SGA from GV\$SQLArea, grouped by SQL-ID, no_plsql=true

	SQL-ID	SQL-Text	C	V	P	Last active	User	Parse	Execs	Elapsed	Ela/Ex.	CPU	Disk Reads	Disk/Ex.	Buffer Gets	B
1	9d9maqkujj39x	select * from table(oman.om_mo_baur.getA	2	2	1	2024-01-04 16:21:11	BAUR_APPU:	BAUR_APPU:	225,930	213,051	0.943	198,991	944,964,912	4,183	3,766,835,518	
1	dmfrg819hc9u	SELECT /*+ "Package journal.OnlineFWTool	1	1	1	2024-01-04 16:21:12	JOURNAL	JOURNAL	345,743,926	146,026	0.000	48,005	2,278	0	1,451,805,049	

Statement details of current SGA from GV\$SQL: Instance = 1, SQL-ID = 'dmfrg819hc9u', Child-No. = 1

/* single line SQL-text formatted by Panorama */
SELECT /*+ "Package journal.OnlineFWTools" */ ID, ID_OFMESSAGE
TYPE,
PROCESSSTART
FROM JOURNAL.OFBULKGROUP
WHERE ROWID = :B1 FOR UPDATE NOWAIT

Parsing schema name	JOURNAL	
Plan-Hash-Value / Opt.Env.	3888655144 / 3016386464	
Parsing module / action	ONLINEFRAMEWORK / Get Next Bulkgroup	
PL/SQL program / line	JOURNAL.ONLINEFWTOOLS : 248	
First Load Time	2023-12-14 05:22:05	
Last Load Time	2023-12-30 20:39:15	
Last Active Time	2024-01-04 16:21:18	
Buffer cache hit ratio	100.00 %	
Metric-Name	Total	Per Exec.
Executions	345,744,225	-
Parse calls	25,988	0.00
Fetches	193,153,087	0.56
Sorts	0	0.00
Rows processed	193,117,751	0.56
Buffer gets	1,451,806,140	4.20
Disk reads	2,278	0.00
Elapsed Time (sec), incl. parallel Query Slaves	146,026	0.00
CPU-Time (sec)	48,006	0.00
Application Wait Time (sec)	1	0.00
Concurrency Wait Time (sec)	96,433	0.00
Cluster Wait Time (sec)	5,084	0.00
User-IO Wait Time (sec)	0	0.00
PL/SQL Exec Time (sec)	0	0.00

Execution-Plan (1)

Bind variables (1)

Objects (1)

Full history

Cursor Sharing (1 versions)

Active Session History

Open Cursor (96)

Expand

DBMS_XPLAN

SQL-Monitor (0)

SQL-Translation

SQL-Patch

Evaluation of the SQLs currently in SGA as well as the history from AWR

Listing of SQLs sorted by various criterias

Details per SQL incl.

- Execution plan
- Bind variables
- Child cursors
- Full history of all AWR snapshots of SQL
- Reasons for multiple cursors
- SQL monitor recordings
- Create baseline / SQL patch

SQL statement details: Execution plan

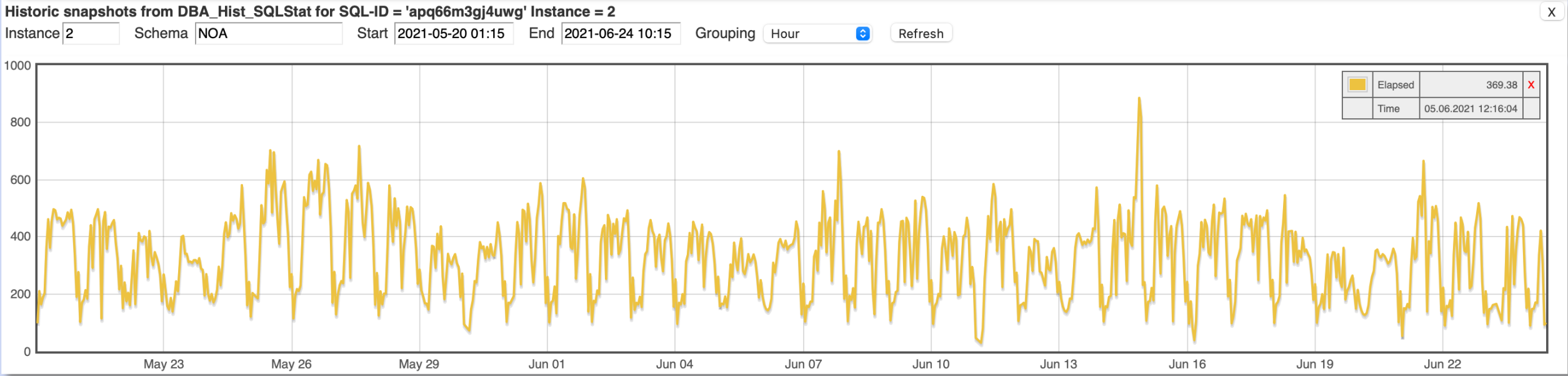
Explain Plan of Child=13 parsed at 02.05.2021 09:53:19, Plan_Hash_Value = 4146904004, Optimizer-Mode=ALL_ROWS																				First ASH-Sample in SGA is from 06.05.2021 09:58:34	
Operation	ID	R.	Object-name	Rows	MB	Cost	Card.	Parallel	Access	Filter	Temp est.	Temp max.	DB time	CPU	Waits	I/O	IC	PGA max.	Proj.		
SELECT STATEMENT	0	41				1.165.531															
HASH UNIQUE	1	40				1.165.531	1	Parallelization (from Other_Tag)				0	0,3	1,1		0,5	0,0	60	"MS".		
NESTED LOOPS SEMI	2	39				1.165.530	1					0	0,5	1,7		0,6	0,0	61	"MS".		
HASH JOIN RIGHT SEMI	3	36				1.111.661	17.221		"IO"."ID_STATEATTRIBUTEQUALIT			0	0,2	0,6		0,2	0,0	60	(#key		
TABLE ACCESS BY INDEX ROWID BATCHED	4	2	SYSPEKS.STATEATTRIBUTEQUALITY	11	0	2	10												"IO.Σ		
INDEX RANGE SCAN	5	1	SYSPEKS.IX_ATTRQUALTY_CODE	11	0	1	10		"IO_SAQ"."CODE">1										"IO.Σ		
HASH JOIN RIGHT SEMI	6	35				1.111.659	17.395		"IO"."ID_PROCESSINGSYSTEMCO			0	0,2	0,6		0,3	0,0	60	(#key		
VIEW	7	7	SYSPEKS.INDEX\$_JOIN\$_012			1	3			("IO_PROC"."CODE" = 5 OR									(rows		
HASH JOIN	8	6							ROWID = ROWID										(#key		
INLIST ITERATOR	9	4																	ROW		
INDEX UNIQUE SCAN	10	3	SYSPEKS.IX_PROCSYSCOD_LPKEY	14	0	0	3		("IO_PROC"."CODE" = 5 OR										ROW		
INDEX FAST FULL SCAN	11	5	SYSPEKS.PKEY_PROCSYSCOD	14	0	1	3												ROW		
NESTED LOOPS	12	34				1.111.658	52.191												"MS".		
NESTED LOOPS	13	32				1.111.658	169.288					0	0,2	0,6		0,3	0,0	60	"MS".		
HASH JOIN SEMI	14	30				1.034.872	24.184		"IC"."ID_COUNTRY" = "C"."ID" AND		23	0	1,5	5,7		2,8	0,2	61	(#key		
HASH JOIN	15	28				981.763	210.867		"MS"."ID" =		19								(#key		
HASH JOIN RIGHT SEMI	16	26				962.611	211.434		"IEP"."ID_SHOPDOMAIN" =		17								(#key		
TABLE ACCESS FULL	17	8	BUYING.ITEMPERMISSION	76.783.942	3.840	134.796	738.307			("IP"."ID_SHOPDOMAIN" = 83		0	20,1	9,2	24,0	6,8	5,1	14	(rows		
HASH JOIN RIGHT SEMI	18	25				823.735	811.492		"I"."ID_STATEATTRIBUTEQUALITY"										(#key		
TABLE ACCESS BY INDEX ROWID BATCHED	19	10	SYSPEKS.STATEATTRIBUTEQUALITY	11	0	2	10												"I_SA		
INDEX RANGE SCAN	20	9	SYSPEKS.IX_ATTRQUALTY_CODE	11	0	1	10		"I_SAQ"."CODE">1										"I_SA		
HASH JOIN	21	24				823.727	819.689		"IE"."ID_ITEM" = "I"."ID"		45								(#key		
HASH JOIN RIGHT SEMI	22	22				759.174	819.690		"IE"."ID_PROCESSINGSYSTEMCO										(#key		
VIEW	23	15	SYSPEKS.INDEX\$_JOIN\$_016			1	3			("IE_PROC"."CODE" = 5 OR									(rows		
HASH JOIN	24	14							ROWID = ROWID										(#key		
INLIST ITERATOR	25	12																	ROW		
INDEX UNIQUE SCAN	26	11	SYSPEKS.IX_PROCSYSCOD_LPKEY	14	0	0	3		("IE_PROC"."CODE" = 5 OR										ROW		
INDEX FAST FULL SCAN	27	13	SYSPEKS.PKEY_PROCSYSCOD	14	0	1	3												ROW		
HASH JOIN	28	21				759.153	3.005.528		"IEP"."ID_ITEMEFFORT" = "IE"."ID"		92	0	2,4	9,2		0,1	0,1	68	(#key		
NESTED LOOPS	29	19				510.319	3.005.528												"C"."I		
TABLE ACCESS BY INDEX ROWID	30	17	SYSP.COUNTRY	248	0	1	1												"C"."I		
INDEX UNIQUE SCAN	31	16	SYSP.IND_COUNTRY_CODEA2_UNIQUE	248	0	0	1		"C"."CODEA2" = 'GB'	UPPER("CODEA2") = 'GB'									"C".R		
TABLE ACCESS FULL	32	18	BUYING.ITEMEFFORTPERMISSION	312.574.92	14.528	510.318	3.005.528			("IEP"."ID_SHOPDOMAIN" = 83		0	30,9	33,9	29,8	27,4	87,7	40	"IEP"		
INDEX FAST FULL SCAN	33	20	BUYING.IX_ITEMEFFORT_BSKAT	85.175.000	5.307	180.601	43.031.10			"IE"."ID_STATEDELETED" = 1		0	3,5	13,2		0,3	2,5	68	"IE".		
TABLE ACCESS FULL	34	23	BUYING.ITEM	5.674.191	1.664	58.232	2.606.693			("I"."ID_STATEATTRIBUTEQUALIT		0	1,2	4,6		0,0	0,0	84	(rows		

SQL statement details: AWR history

Historic snapshots from DBA_Hist_SQLStat for SQL-ID = 'apq66m3gj4uwg' Instance = 2

Instance2SchemaNOAStart2021-05-20 01:15End2021-06-24 10:15GroupingHourRefresh

Start	First occurrence	Last occurrence	Plans	Plan hash value	Envs.	Opt. env hash value	Execs	Elapsed	Ela. / Ex.	Ela. / Row	CPU	App.	Con.	Clu.	I/O	PL/SQL	Disk Reads	Disk / Ex.	ms / Disk Read	Hit Ratio	Buffer Gets	Buffer / Ex.	Buffer / Row	Rows proc.	Rows / Ex.	Parses
2021-06-24 10:00:00	2021-06-24 10:00:07	2021-06-24 10:15:03	1	2238303688	1	1683692210	421,968	100	0.0002	0	50	0	1	0	23	24	60,196	0.143	0.37	97.14	2,104,214	4.99	5.06	416,068	0.99	0
2021-06-24 09:00:00	2021-06-24 09:45:31	2021-06-24 10:00:07	1	2238303688	1	1683692210	410,333	93	0.0002	0	49	0	1	0	18	23	58,468	0.142	0.31	97.14	2,046,823	4.99	5.05	405,085	0.99	1
2021-06-24 08:00:00	2021-06-24 08:00:19	2021-06-24 09:00:16	1	2238303688	1	1683692210	1,279,603	315	0.0002	0	166	0	2	0	70	75	210,311	0.164	0.33	96.71	6,386,473	4.99	5.04	1,267,592	0.99	15
2021-06-24 07:00:00	2021-06-24 07:00:10	2021-06-24 08:00:19	1	2238303688	1	1683692210	1,775,193	422	0.0002	0	207	0	3	0	103	97	279,368	0.157	0.37	96.85	8,868,559	5.00	5.02	1,767,650	1.00	1



SQL statement details: SQL Monitor

Enterprise Manager Database Active Report

Real-time SQL Monitoring: 0u8czynm2gpar Status ✔

Time Zone Browser (GMT+02:00) ▼

Overview

General

SQL Text

SELECT PCSELEKT.ART.BEST_WG_FT_KZ FT, PCSELEKT....

Execution Started

06.05.2021 16:58:25 GMT+02:00

Last Refresh Time

06.05.2021 17:01:07 GMT+02:00

Execution Id

16777216

User Name

THEINS@EKR30PS

Fetch Calls

1

Time & Wait

Duration

2,7 m

Database Time

2,97 m

PL/SQL & Java

0 s

Activity

100 %

I/O

Buffer Gets

1.009K

I/O Requests

334K

I/O Bytes

2,6 GB

Details

Plan Statistics

SQL Text

Activity

Metrics

Plan Hash Value 1020049330 Plan Size 20 Lines

Plan Notes

Operation	Object	Information	Line ID	Timeline	Execut	Est. Rows	Rows	Mem (Max)	Temp (Max)	I/O Requests	Activity
SELECT STATEMENT			0		1						
SORT ORDER BY			1		1	18K					
HASH GROUP BY			2		1	18K					
FILTER			3		1						
HASH JOIN OUTER			4		1	18K	1.552	1,8 MB			5,03 %
FILTER			5		1		1.552				
HASH JOIN OUTER			6		1	18K	1.552	1,8 MB			3,77 %
NESTED LOOPS			7		1	18K	1.552				
NESTED LOOPS			8		1	55K	1.552				
HASH JOIN			9		1	55K	1.552	2,6 MB			0,63 %
TABLE ACCESS BY INDEX ROWID BATCHED	ART		10		1	35K	1.008			3	
INDEX RANGE SCAN	IND_ART_6		11		1	20K	2.000			12	
INLIST ITERATOR			12		1		29M				

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Sessions currently connected to DB

Menu „DBA general“ / „Sessions“

Show sessions

Active only ☒ Parallel Query ☐ User only ☒ DB links only ☐ Inst. Filter

Show sessions

Sessions

I	SID/SN	Status	SQL_ID	Wait event	Wait time	User	Proc	Machine	C-User	C-Proc	Program	Client-Info	Client ID	Module	Action	Service	Logon Time	Last Active Start	I/O-Index
2	5649, 25284	ACTIVE		ON CPU	0.0	NOA	96382	coreprod025	core	1234	JDBC Thin Client			OFMSGType_	process OFMessages	NOADB_LOG	2021-06-22 09:29:48	2021-06-24 09:34:24	556,476
2	5676, 52716	ACTIVE	7xbj30xa9v971	SQL*Net message from client	0.0	NOA	319141	coreprod016	core	1234	JDBC Thin Client	ApplExecID = 11058136 WsMethodIC = 236		ID_Application = 128	ID_WSMetho = 236	NOADB_ONLI	2021-06-20 00:45:51	2021-06-24 09:34:24	776,876

Details for session SID=5649, Serial#=25284, Instance=2, 2021-06-24 09:34:55

Status	Client-Info	Module	-Action	Username	Audit-SID	Sh. Proc.	PID	Machine	OSUser	Process	Program	Logon-time	Last active start	Tx. ID	Auth. type	Client char set	Client connection	Client OCI lib	Client version
ACTIVE		OFMSGType_	process OFMessage	NOA	327340	96382	3143	coreprod02	core	1234	JDBC Thin Client	2021-06-22 09:29:48	2021-06-24 09:34:54	6E00020012EAF06	DATABASE	Unknow	Heterogeneous	Unknow	12.1.0.2.0

	SQL-ID	C.	SQL exec start	SQL exec ID	SQL-Text
Aktuelles SQL-Statement	gp109ra3nwca1	1	2021-06-24 09:34:53	42,927,715	BEGIN AUFTRAG.AU_SE_FKT_NOA_FAKTURA_LW_JL.fakturiere(:1 ,
Vorheriges SQL-Statement	9anb1gf84m4tm	44	2021-06-24 09:34:53	47,947,422	INSERT /** APPEND "UT_MO_BUSINESS_LOG.schreibe_log" */

Process Memory (incl. PQ-Server) from GV\$Process_Memory

Category	Allocated	Used	Max Allocated
Freeable	2,162,688	0	
PL/SQL	9,350,864	737,008	23,969,056
Other	19,920,268		19,920,268
SQL	3,528,936	38,960	7,541,472

Wait-Status

Locks

Temp-Usage

612 open cursor

Objects accessed

Active Session History

Session-Statistics

Audit Trail

Optimizer Env.

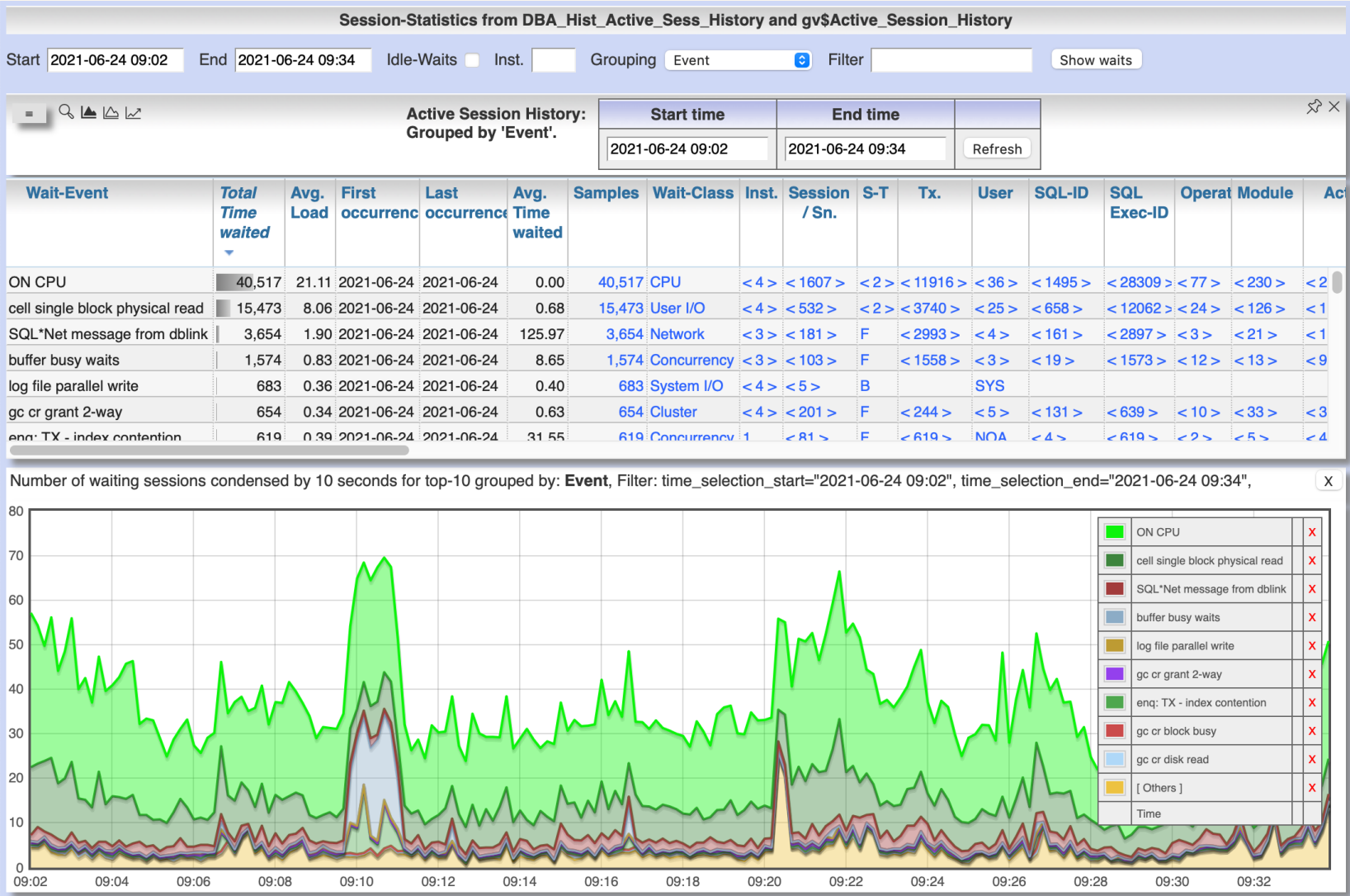
SQL-Monitor (77)

Listing of DB sessions with optional filter and drilldown into session and SQL details

Filter and sort sessions by several criterias

Active Session History

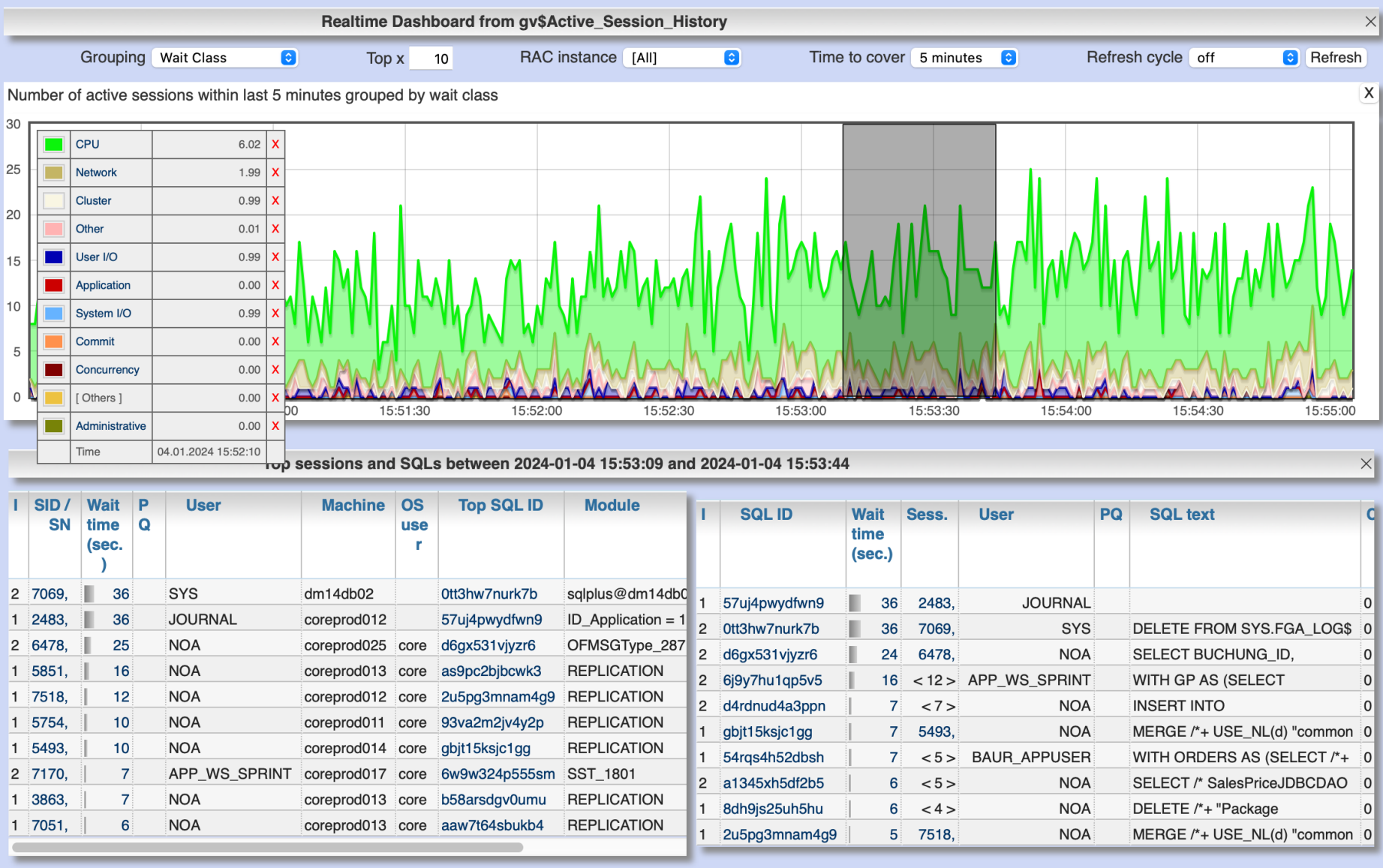
Menu „Analyses/statistics“ / „Session waits“ / „Historic“



- Select period and entry grouping
- Successive drilldown according to various criteria
- „< xxx>“ shows number of different values of the category column in current selection
- Click on „< xxx>“ groups the values of the current row according to this category in another table
- Drill down to individual ASH sample records

ASH: Real-time dashboard

Menu „DBA general“ / „Dashboard“



- Real-time monitoring of active sessions grouped by several criterias (like „wait class“ in picture)
- Automatic refresh in desired cycle
- Top SQLs and Top sessions for choosen period
- Drilldown by wait class (click in legend), session or SQL

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Structure info for tables, indexes, packages etc.

Menu „Schema/Storage“ / „Describe object“

Describe database object

Object-Owner

cust

Object-Name

Customer

Object-Type

[Alle]

Describe

Description of TABLE CUST.CUSTOMER

Company dependent customer

Komprimierung reduziert Footprint auf 60%

Columns of TABLE CUST.CUSTOMER

Col.Name	Type	Prec.	Sc.	N.	Def.	Distinct	Nulls	Avg. Len.	Density	Buckets	Histogram	Comments	LOB segment	EQ	EQJ	NEQ
ID	NUMBER	9	0	N		68.305.085	0	7	0,0000	1	NONE	Primary Key		24.729	84.737	0
ID_COMPANY	NUMBER	4	0	N		59	0	3	0,0000	59	FREQUENCY	Company to		48.698	37.917	0
ID_PERSON	NUMBER	9	0	N		68.305.085	0	7	0,0000	1	NONE	Person that		9.015	26.073	0
CREATIONDATE	DATE			N		43.819.008	0	8	0,0000	254	HYBRID	Date of		5.626	192	78
ID_LOCALE	NUMBER	4	0	N		13	0	3	0,0000	13	FREQUENCY	Locale		7.318	4.971	0
CUSTNO	VARCHAR2	10 Chars		N		39.424.000	0	10	0,0000	254	HYBRID	Company		12.343	1.714	0
CANCELLATIONDATE	DATE			Y		96.240	64.013.335	2	0,0000	254	HYBRID	Date when		7.322	0	0
ORDERCOUNTER	NUMBER	6	0	Y		2.533	12.500.973	3	0,0000	254	TOP-	Comments		119	0	0
LASTONLINEUPD	DATE			Y		46.915.584	7.174.628	8	0,0000	1	NONE	Letzte				
ID_SHOPDOMAIN	NUMBER	4	0	Y		88	44.031.316	3	0,0114	1	NONE	ShopDomain,				

Attributes of TABLE CUST.CUSTOMER

TS	Pct Free	Init. Trans	Initial extent (KB)	Rows	Size (MB) Table	Size (MB) Total	Extents	Blocks	Empty	Avg. Space	Chained	RowLen	Dg.	Cache	Part.	Sub-Part.	Created	La
DATA02	10	8		68.305.085	4.659,75	18.598,94	598	596.448	0	0	0	56	1	N	2	0	08.03.2015 00:42:45	05.08

6 Indexes

1 Primary Key

2 Check Constraints

4 References from

85 References to

12 Triggers

129 Dependencies

49 Grants

DBMS_METADATA

DB-Cache

Sessions accessing

SQLs

Active Session History

Segment statistics

Size evolution

Detailed info on DB objects with drilldown into structures, accesses, dependencies, etc.

Storage usage overview

Menu „Schema/Storage“ / „ Disc storage summary“

Storage-objects for database NOADB_PROCESS 2021-06-24 10:53

Sums total

Typ	MB Total	MB Used	MB Free	% used
PERMANENT	11,168,763	8,596,727	2,572,036	77.0
TEMPORARY	956,121	873,918	82,203	91.4
UNDO	917,504	789,474	128,030	86.0
Redo-Logs outside FRA	196,608	196,608	0	100.0
Fast Recovery Area	9,216,000	5,122,253	4,093,747	55.6
TOTAL	22,454,996	15,578,980	6,876,016	69.4

Net sums in TS by segment types

Segment-Type	MBytes
TABLE	4,637,933
INDEX	3,677,001
TYPE2 UNDO	789,465
LOBSEGMENT	
SYSTEM STATISTICS	
CLUSTER	

Usage of fast recovery area (FRA)

total FRA size = 9,000.00 GB

not reclaimable usage = 34.77 %

File type	MB used	% used	% recl.
	0.00	50.00	20.81
	2.80	5.58	0.00

Size of segment type (used space net) in MBytes

= 3,590.8 Gigabytes

= 3.5 Terabytes

= 39.18 % of column sum 9386024

Schema-Usage

Schema	Table	Index	Index IOT-PKey	TYPE2 UNDO	SYSTEM STATISTICS	ROLLBACK	Total MB
AUFTRAG	1,984,192	2,078,643	26,900				4,089,736
CUST	297,175	634,812	605,292				1,537,279
SYS	66,152	35,559	24	789,465	8,447	0	899,646
LOGSTICK	884,888	848,488	0				1,733,376

Tablespace-usage per schema

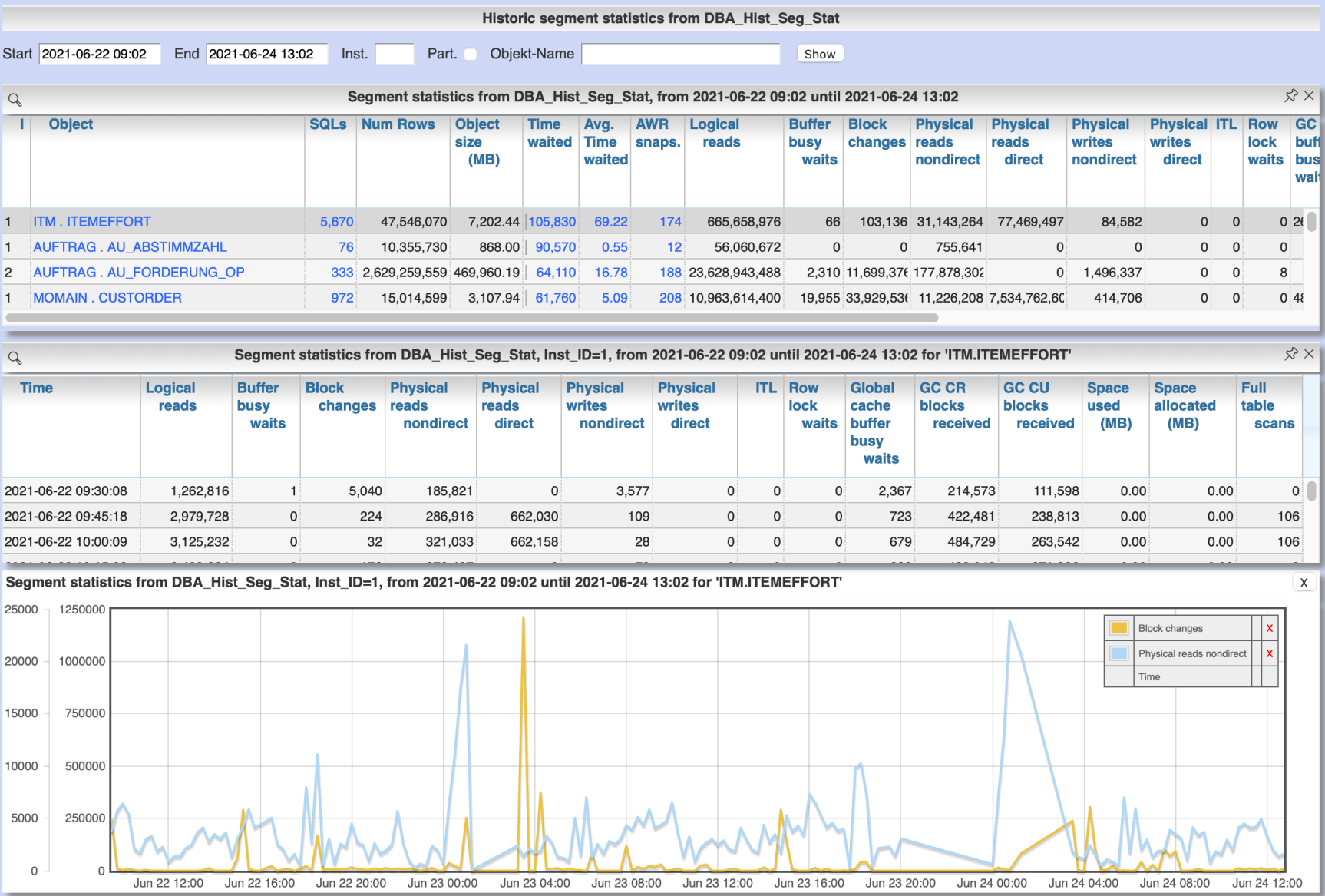
Schema	Tablespace	Used (MB)	Quota (MB)	Charged (MB)
AUFTRAG	DATASP01	1,979,282		
CUST	DATA02	558,223		
CUST	INDEX01	329,599		
CUST	DATA01	317,831		
CUST	INDEX02	278,581		

Tablespace-usage total from DBA_Tablespaces

Tablespace	Contents	Block size	MB Total	MB Used	MB Free	% used	Auto ext.	MB max.	% used max.	Files	Status	Logging	Force log.	Extent mgmt.	Allocation Type	PI in	Segment Space Mgmt.	Def. table compress	Big file	Encryp	Def. in memory
DATASP01	PERMANENT	8,192	2,867,200	2,521,547	345,654	87.9	NO			1	ONLINE	LOGGING	NO	LOCAL	SYSTEM	NO	AUTO	DISABLED	YES	NO	DISAB
INDEXSP01	PERMANENT	8,192	2,867,200	2,497,308	369,892	87.1	NO			1	ONLINE	LOGGING	NO	LOCAL	SYSTEM	NO	AUTO	DISABLED	YES	NO	DISAB
DATA02	PERMANENT	8,192	1,730,560	1,157,123	573,437	66.9	NO			1	ONLINE	LOGGING	NO	LOCAL	SYSTEM	NO	AUTO	DISABLED	YES	NO	DISAB
DATA01	PERMANENT	8,192	1,228,800	962,742	266,058	78.3	NO			1	ONLINE	LOGGING	NO	LOCAL	SYSTEM	NO	AUTO	DISABLED	YES	NO	DISAB
INDEX01	PERMANENT	8,192	921,600	666,775	254,825	72.3	NO			1	ONLINE	LOGGING	NO	LOCAL	SYSTEM	NO	AUTO	DISABLED	YES	NO	DISAB

Segment statistics

Menu „Analyses / statistics“ / „Segment statistics“ / „Historic“



Characteristic values per table/index for period

Sortable according to individual characteristic values

Evolution over time within the period

Thank you for your interest



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